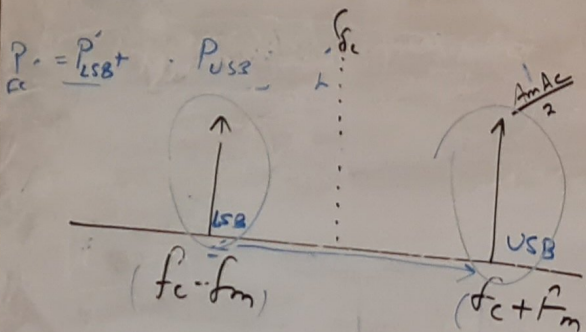


DSB-SC / SSB

DSB-SC (suppressed carrier)



$$BW = 2f_m$$

$f_{max} - f_{min}$

$$\eta = \frac{P_{SB}}{P_t} \times 100\%$$

$$S(t) = m(t) \cdot c(t)$$

DSB-SC

$$S(t) = [A_m \cos 2\pi f_m t \cdot A_c \cos 2\pi f_c t]$$

DSB-SC

$$S(t) = A_m A_c \cos 2\pi f_m t \cos 2\pi f_c t$$

DSB-SC

general equation of DSB-SC

$$P_t = P_{USB} + P_{LSB}$$

$$= \frac{A_m^2 A_c^2}{8R} + \frac{A_m^2 A_c^2}{8R} = \frac{A_m^2 A_c^2}{4R}$$

$$P_t = 2 \cdot P_{USB/LSB} = P_{SB}$$

$$\eta = \frac{P_{SB}}{P_t} \times 100\%$$

$$\eta = 100\%$$

$$\eta_{Fc} = \frac{P_{SB}}{P_t}$$

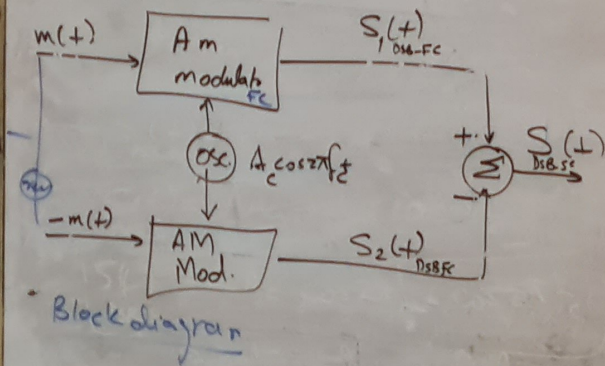
$P_{SB} + P_c$

DSB-SC / SSB

DSB-SC (suppressed Carrier)

1. Modulator or

Balanced Modulator.



$$S_1(t) = A_c [1 + K_a m(t)] \cos 2\pi f_c t$$

$$S_2(t) = A_c [1 - K_a m(t)] \cos 2\pi f_c t$$

$$S(t) = S_1(t) - S_2(t) = A_c [1 + K_a m(t)] \cos 2\pi f_c t - A_c [1 - K_a m(t)] \cos 2\pi f_c t$$

$$S(t) = A_c \cos 2\pi f_c t + A_c K_a m(t) \cos 2\pi f_c t - A_c \cos 2\pi f_c t + A_c K_a m(t) \cos 2\pi f_c t$$

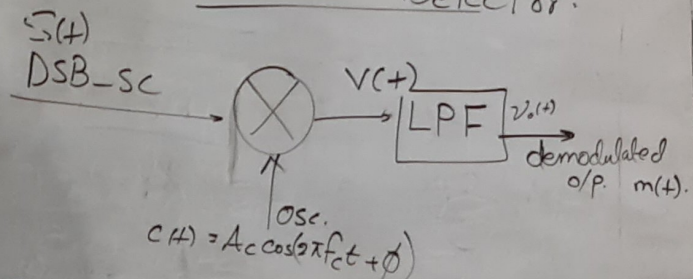
$$S(t) = 2 A_c K_a m(t) \cos 2\pi f_c t = \underbrace{2 K_a}_{\text{scalar coeff.}} \cdot \underbrace{m(t)}_{\text{message}} \cdot \underbrace{c(t)}_{\text{carrier}} \rightarrow \boxed{S(t) = m(t) \cdot c(t)}_{\text{DSB-SC}}$$

DSB-SC / SSB

DSB-SC (suppressed Carrier)

De-Modulator

1 - Coherent Detector.



$$S(t) = A_c \cos 2\pi f_c(t) \cdot m(t)$$

$$V(t) = S(t) \cdot C(t)$$

$$v(t) = A_c \cos(2\pi f_c t) m(t) \cdot A_c \cos(2\pi f_c t + \phi)$$

$$= A_c^2 m(t) [\cos 2\pi f_c t \cdot \cos(2\pi f_c t + \phi)]$$

$$v(t) = \frac{A_c^2}{2} [\cos \phi m(t) + \cos(4\pi f_c t + \phi) m(t)]$$

$$v_o(t) = \text{LPF}_{f_m}[v(t)]$$

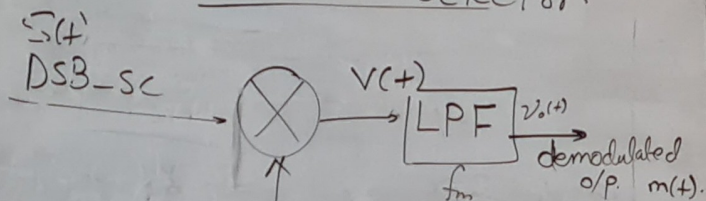
$$v_o(t) = \frac{A_c^2}{2} \cos \phi m(t)$$

* Problem $\rightarrow$ quadrature null effect
 $\phi = 90^\circ$

DSB-SC (suppressed Carrier)

De-Modulator

1 - Coherent Detector.



$$C(t) = A_c \cos(2\pi f_c t + \phi)$$

$$S(t) = A_c \cos 2\pi f_c t \cdot m(t)$$

$$V(t) = S(t) \cdot C(t)$$

DSB-SC / SSB

$$v(t) = \underbrace{A_c \cos(2\pi f_c t)}_{\text{carrier}} \cdot \underbrace{m(t)}_{\text{message}} \cdot \underbrace{A_c \cos(2\pi f_c t + \phi)}_{\text{carrier}}$$

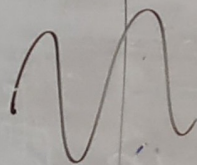
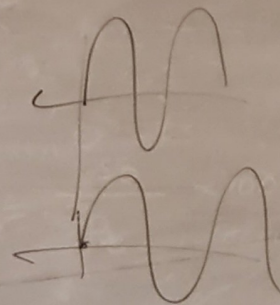
$$= A_c^2 m(t) [\cos 2\pi f_c t \cdot \cos(2\pi f_c t + \phi)]$$

$$v(t) = \frac{A_c^2}{2} \left[\underbrace{\cos \phi}_{\text{cos } 2\pi f_m t} m(t) + \underbrace{\cos(4\pi f_c t + \phi)}_{\text{high frequency}} m(t) \right]$$

$$v_0(t) = \text{LPF}_{f_m} [v(t)]$$

$$v_0(t) = \frac{A_c^2}{2} \cos \phi m(t)$$

* Problem → quadrature null effect
($\phi \pm 90^\circ$)

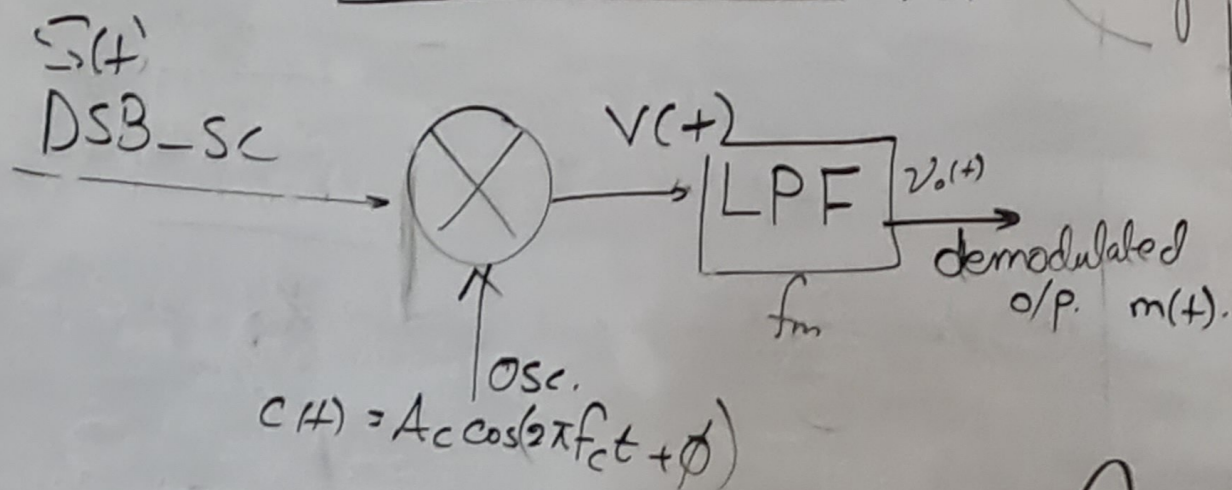


DSB-SC (suppressed carrier) SC.

De-Modulation

1 - Coherent Detector.

Synchronization
between
Tx & Rx
Carrier
Signal



$$v(t) = A_c \cos(2\pi f_c t) m(t)$$

$$= A_c^2 m(t) [\cos \phi]$$

$$v(t) = \frac{A_c^2}{2} [\cos \phi]$$

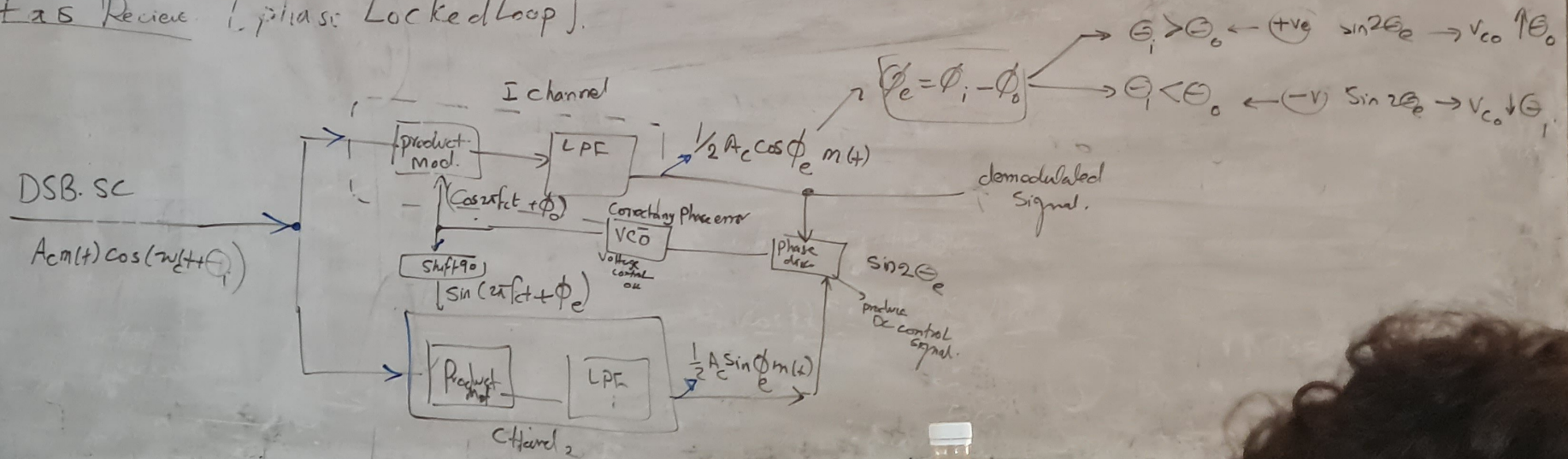
$$v_o(t) = \text{LPF}_{f_m} [v(t)]$$

DSB-SC / SSB

DSB-SC (suppressed carrier) SC.

Demodulation

2. Costas Receiver [Phase Locked Loop]



DSB-SC / SSB

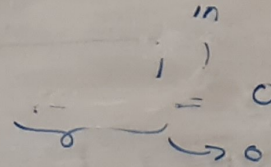
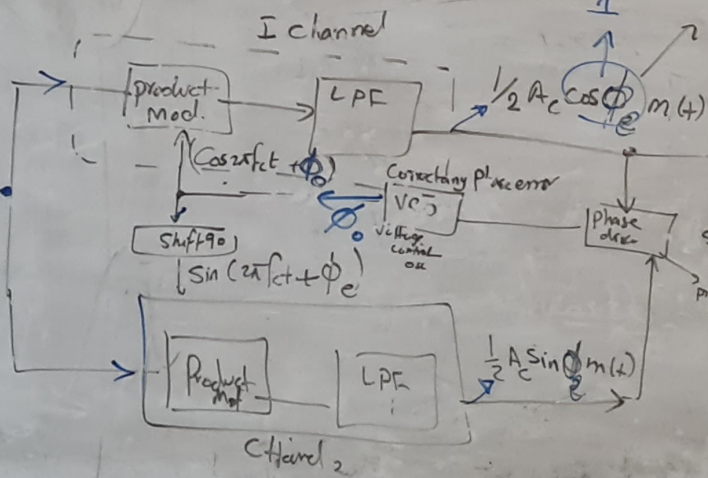
DSB-SC (suppressed carrier) SC.

De-Modulation

2. Costas Receiver [Phase Locked Loop].

$S(t)$
DSB-SC

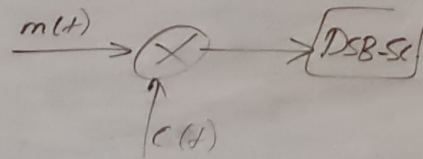
$$A_m(t) \cos(2\pi f_c t + \theta_e)$$



$$\begin{aligned} \theta_e = \phi_i - \phi_o &\begin{cases} \text{+ve} \rightarrow \sin 2\theta_e \rightarrow V_{co} \uparrow \theta_o \\ \text{-ve} \rightarrow \sin 2\theta_e \rightarrow V_{co} \downarrow \theta_o \end{cases} \\ \text{Zero} &\rightarrow \theta_i = \theta_o \end{aligned}$$

DSB-SC / SSB

DSB-SC (suppressed carrier)



a DSB-SC uses a carrier $c(t) = b \cos(2\pi \times 10^6 t)$, Vol:

and an information $m(t) = 3 \cos(2\pi \times 10^4 t)$.

* Remember that $\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$

$$S_{DSB-SC}(t) = c(t) \cdot m(t) = 30 \cos(2\pi \times 10^6 t) \cdot \cos(2\pi \times 10^4 t)$$

$$= 30 \cos(2\pi \times 10^6 t) \cdot \cos(2\pi \times 10^4 t)$$

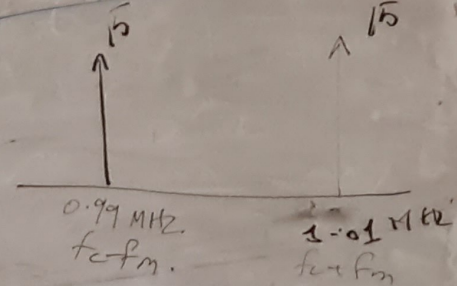
$$= \frac{30}{2} [\cos[2\pi \times 10^6 t + 2\pi \times 10^4 t] + \cos[2\pi \times 10^6 t - 2\pi \times 10^4 t]]$$

2. Bandwidth:

$$B = 2f_m$$

$$Bw = 2 \times 10^4$$

4) draw frequency spectrum



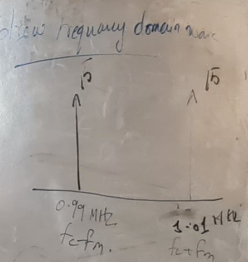
3) Power P_t

$$P_t = \frac{A_c^2 f_m}{4R} \approx 1W$$

$$\eta = \frac{P_{SD}}{P_t} = \frac{100}{100} = 100\%$$

$$P_{DSB} = \frac{P_{SD}}{2} = 50W$$

DSB-SC / SSB



and an information $m(t) = 3 \cos(2\pi \times 10^4 t)$

④. write down a DSB-modulated signal Formula/Expression

$$B_w = 2 \times 10^4$$

$$\begin{aligned} S(t) &= C(t) \cdot m(t) = 30 \cos(2\pi \times 10^6 t) \cdot \cos(2\pi \times 10^4 t) \\ &= 30 \cos(2\pi \times 10^6 t) \cdot \cos(2\pi \times 10^4 t) \\ &= \frac{30}{2} \left[\cos[2\pi \times 10^6 t + \pi \times 10^4 t] + \cos[2\pi \times 10^6 t - \pi \times 10^4 t] \right] \end{aligned}$$

③ Power P_L $\sim \frac{1}{2}$
 $P_L = \frac{I_{ac}^2 R_m}{4R_{in}}$

$$\eta = \frac{P_{SD}}{P_T}$$